function in a neural network trained using back propagation?

9. In a neural network built for classification problems, the output layer typically has neurons whose activation function is a softmax activation function. Can you explain why you should use a softmax function for the output layer of a classification network? *(Clue:* Think about how the softmax function is defined and how it can be interpreted as a probability.)
10. The back-propagation algorithm is the anchor behind the ability to train very large and deep neural networks. Can you explain the intuition behind the back-propagation algorithm? Instead of using back-propagation, I design a method wherein I disturb the weight of one of the arbitrary neurons in the network and measure its impact on the cost function, to determine the gradient of the cost function with respect to that weight. While this approach certainly seems simple compared to the complicated back-propagation algorithm, explain why this is not a scalable approach?
11. Recall the fact that back-propagation is built on the premise of computing the gradient of the cost function with respect to each of the weights, and using the computed gradient to update the weights for the next pass. What are the criteria a reasonable cost function needs to satisfy for back-propagation to be applied?
12. Different cost functions can be applied for a neural network such as the quadratic cost function, the cross entropy cost function, etc. Can you explain why you may prefer a cross-entropy cost function over a quadratic cost function? *(Clue:* Think of the slowdown in learning for a quadratic cost function when a neuron is near saturation.)
13. Continuing the question of cost functions, very often people use a sigmoid activation function for the output layer along with a cross-entropy cost function; or a softmax output layer with log likelihood cost function. Can you use a softmax layer with the cross-entropy cost function? Explain the reasons behind your choice. *(Clue:* Think of interpreting the softmax layer output as a probability distribution.)
14. How can you compare the performance of two different machine learning algorithms? For example, consider that you have been given two learners – one based on support vector machines and another that uses neural networks for classifying documents. Let us also assume that the SVM based learner performs better than the neural network based learner when the training data size is M, whereas the neural network performs better than SVM when the training data size is much less than M, but is poorer in performance by a few points when comparing training data sizes of M and above. Which training algorithm would you prefer in that case, and can you explain the rationale behind your choice?
15. When we discussed neural networks in our previous column, I did not go into much detail about the hyper-parameters of the learning framework such as the learning rate, number of mini-epochs, etc. Can you explain some of these hyper-parameters and how they impact the classification test accuracy? How would you choose the appropriate values for the hyper-parameters?

One of the best resources for learning about the basic concepts of neural networks is the free online book from Michael Nielsen available at *http://neuralnetworksanddeeplearning.com/index.html.* I would suggest that our readers go through this short book before they attempt to read through the voluminous deep learning text book from Prof. Bengio et al, available at *http://www.deeplearningbook.org/.* I like Nielsen's book, especially because he has provided short Python code snippets for different concepts and you can try them out quickly as you read through the chapters. Another great resource to get familiar with deep learning concepts is the repository of articles and tutorials available at *https://deeplearning4j.org/tutorials.*

Given the enormous number of papers and articles available in this space, it can be very confusing for a novice computer science student who wants to gain expertise in deep learning. While there are online courses available from Coursera (Prof. Hinton's course on neural networks) and Udacity (a deep learning nano-degree), I would suggest that you also read through one of the above-mentioned text books in order to get the mathematical concepts clear, instead of just viewing neural networks as a black-box machine learning algorithm. This would allow you to understand and appreciate the application of different neural networks to different problems. For instance, you may want to dig deeper into why convolutional networks, which were originally envisaged for image recognition, are also useful in text processing tasks such as sentence classification or document classification. More importantly, as you use some of the standard neural net architectures for specific text processing problems such as sentiment analysis or topic modelling, you will understand how different architectures need to be tuned for different tasks. We will discuss more about this in our next column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at *sandyasm_AT_yahoo_DOT_com*. Till we meet again next month, wishing all our readers a wonderful and productive month ahead! END

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